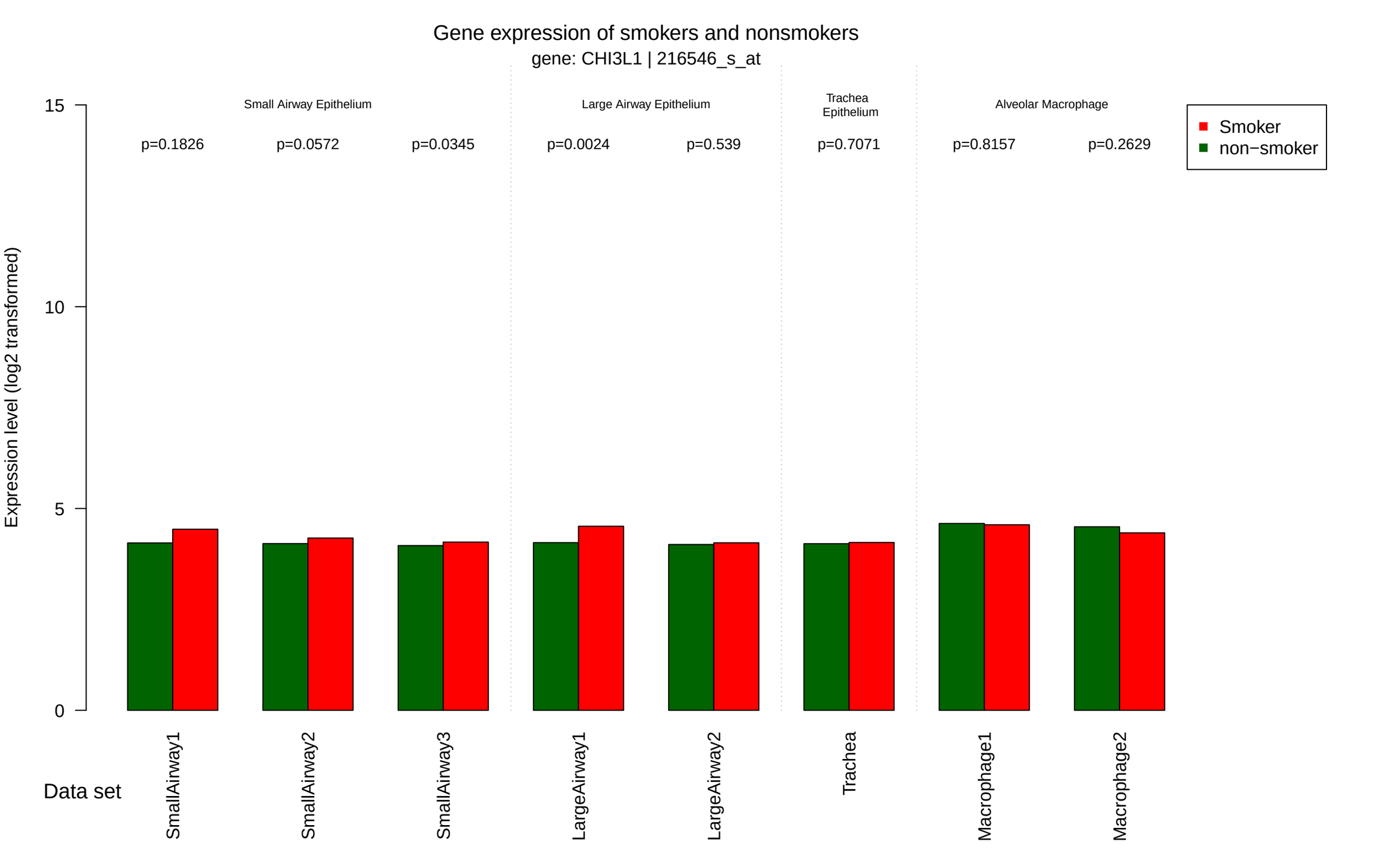
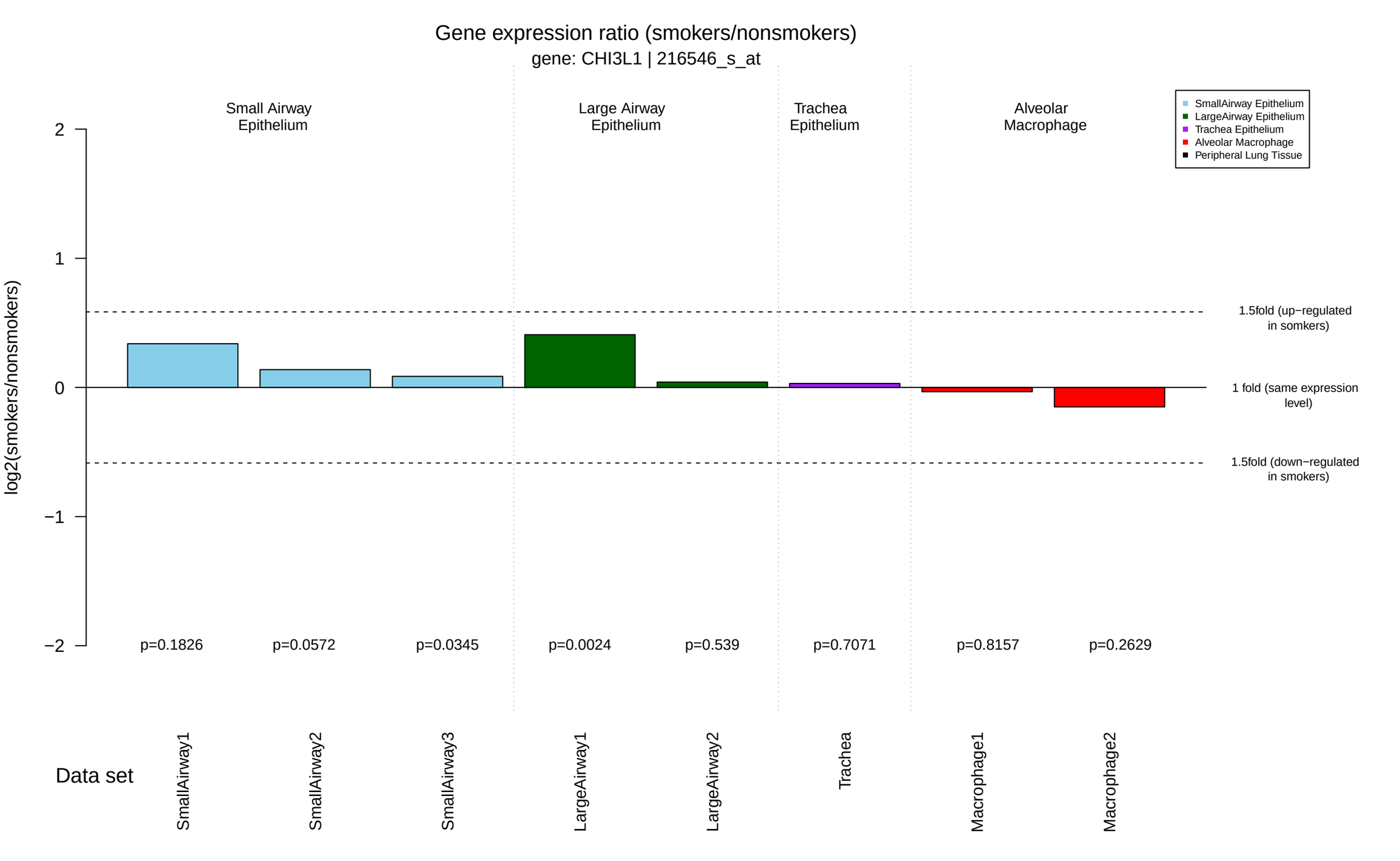
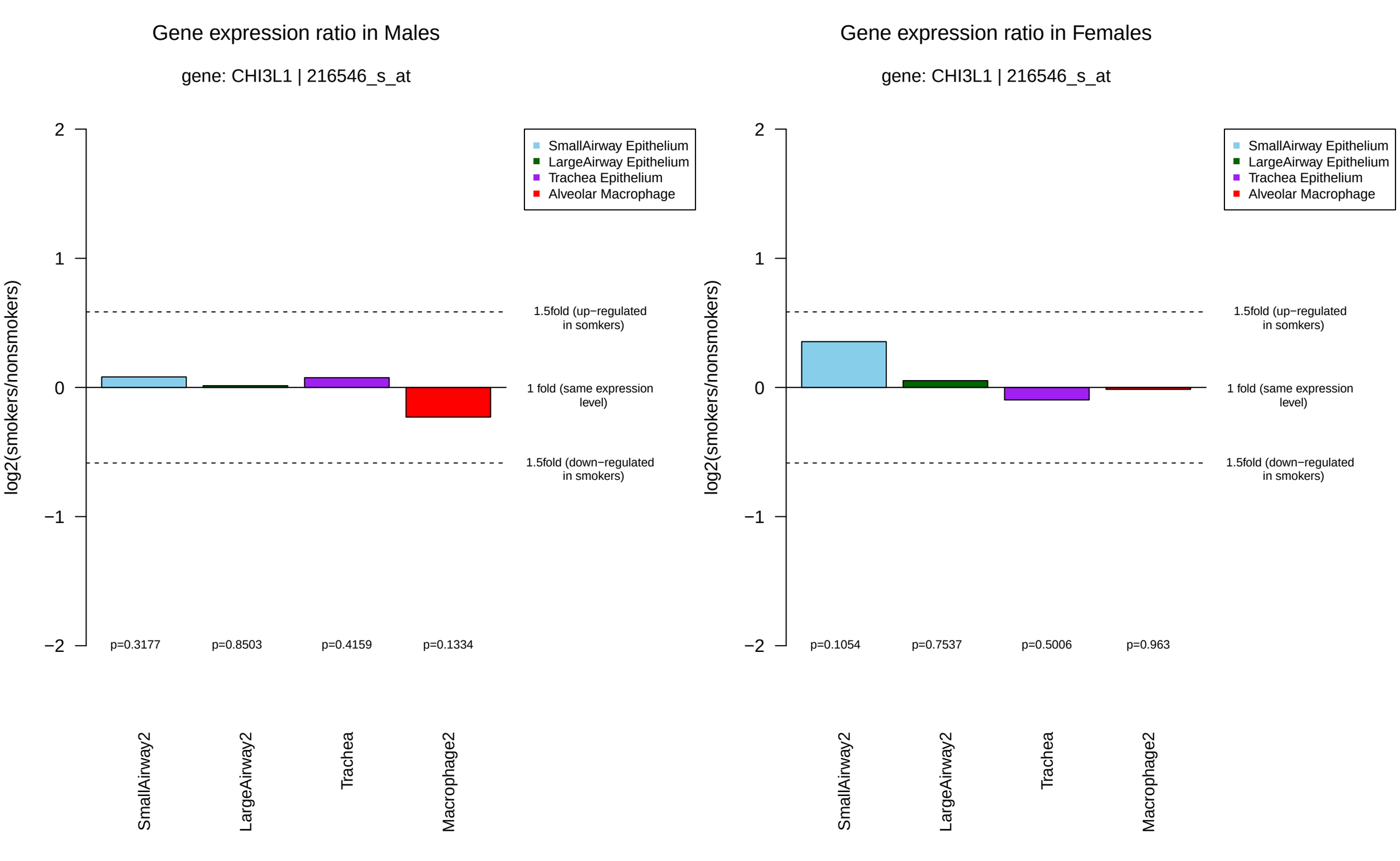
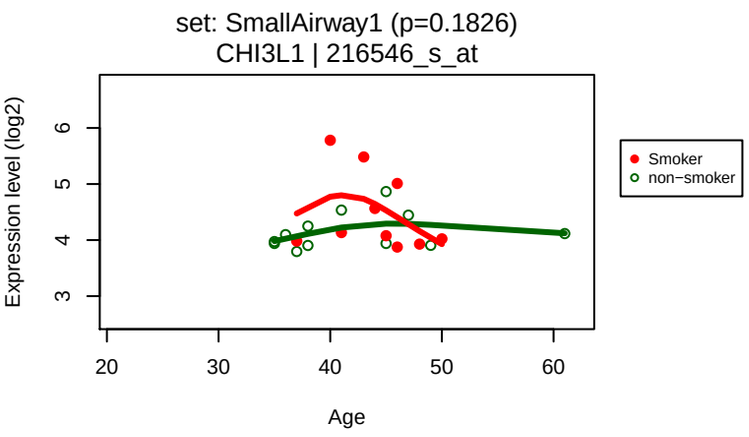
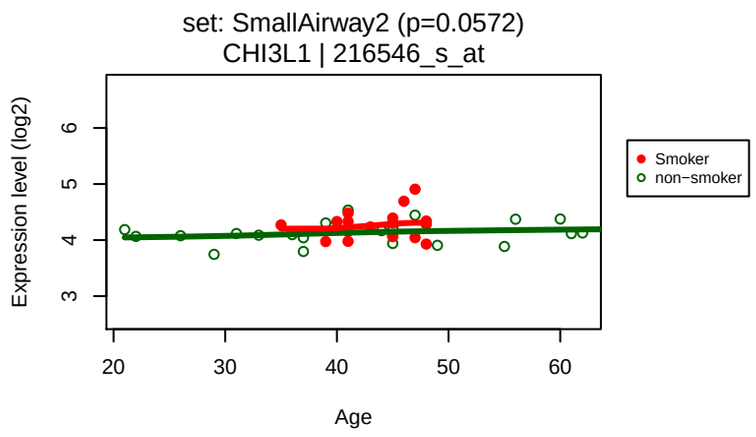
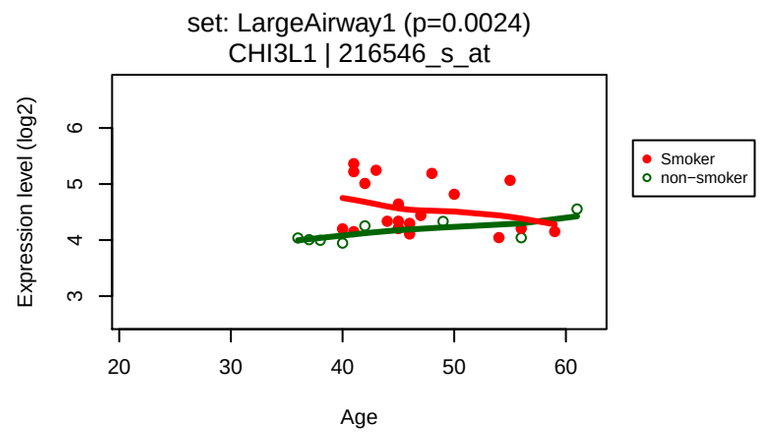

Gene expression of smokers and nonsmokers
gene: CHI3L1 | 216546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1826
p=0.0572
p=0.0345
p=0.0024
p=0.539
p=0.7071
p=0.8157
p=0.2629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

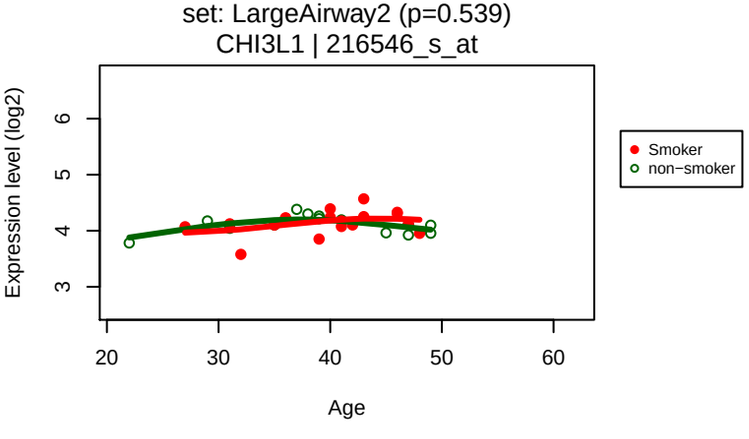

Gene expression ratio (smokers/nonsmokers)
gene: CHI3L1 | 216546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1826
p=0.0572
p=0.0345
p=0.0024
p=0.539
p=0.7071
p=0.8157
p=0.2629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

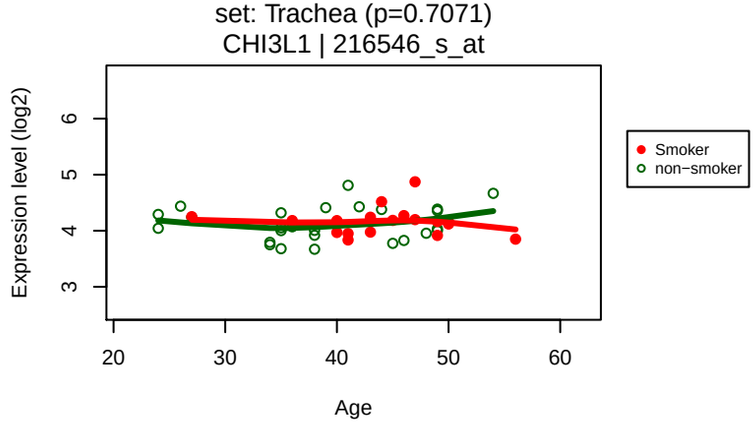

Gene expression ratio in Males
gene: CHI3L1 | 216546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3177
p=0.8503
p=0.4159
p=0.1334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHI3L1 | 216546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1054
p=0.7537
p=0.5006
p=0.963
SmallAirway2
LargeAirway2
Trachea
Macrophage2

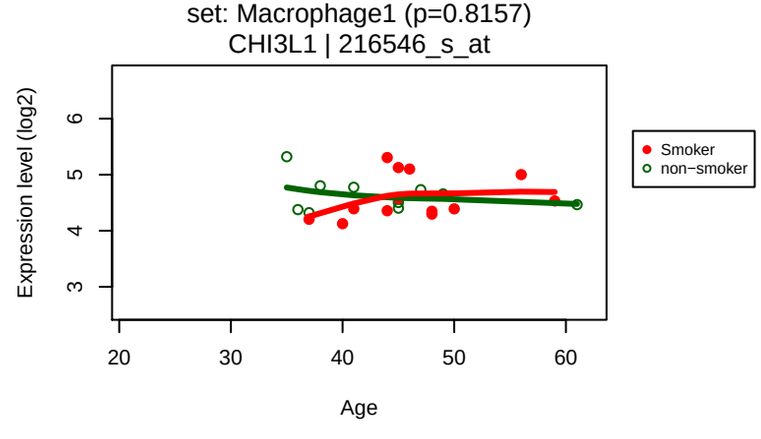

set: SmallAirway1 (p=0.1826)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker

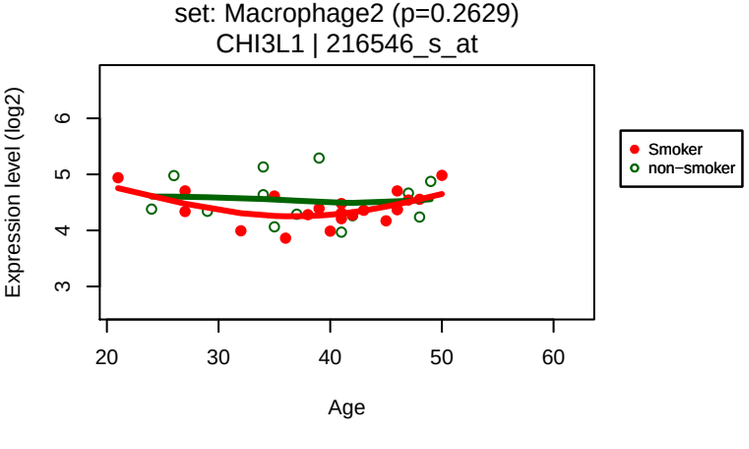

set: SmallAirway2 (p=0.0572)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker

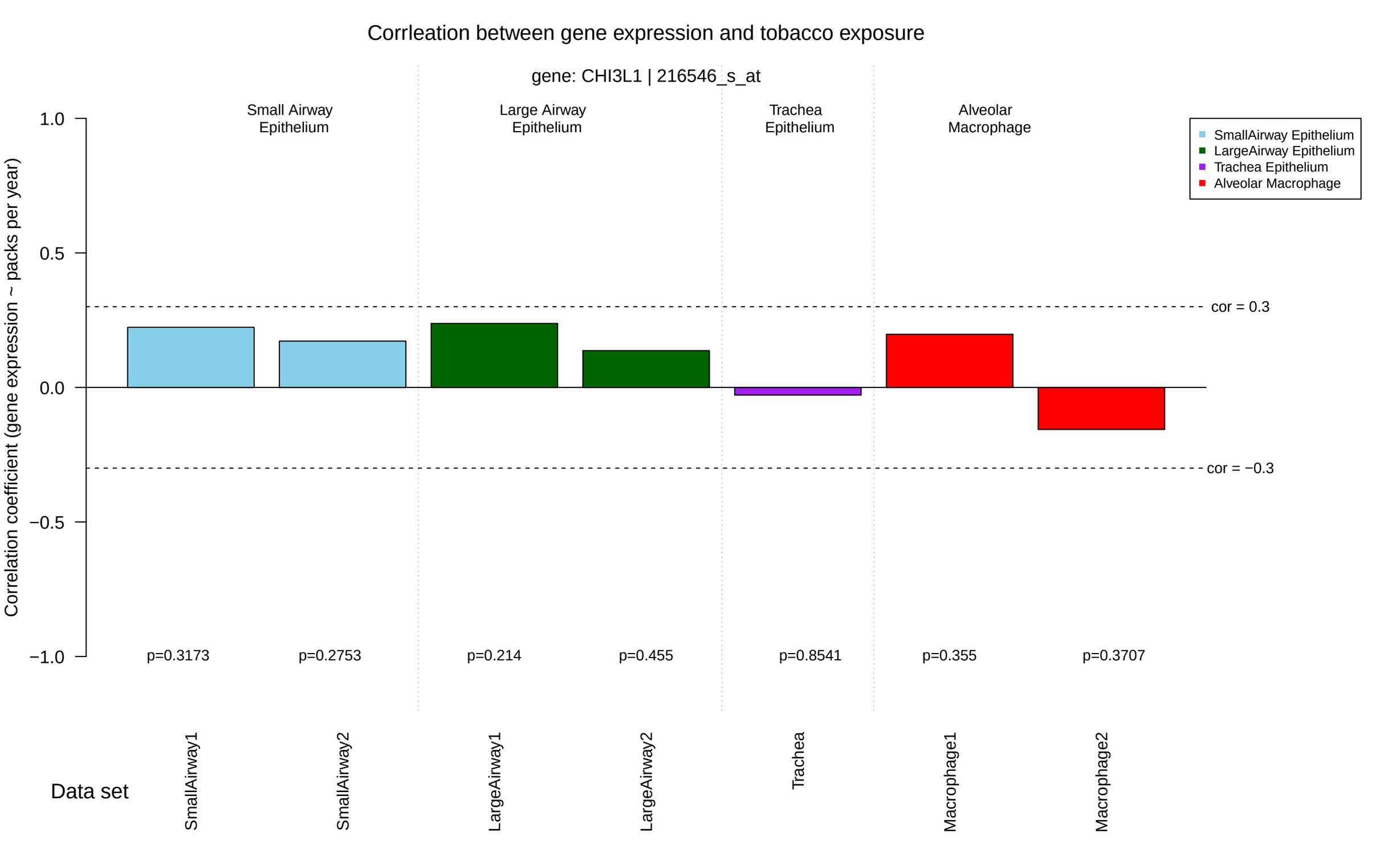

set: LargeAirway1 (p=0.0024)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.539)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7071)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8157)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2629)
CHI3L1 | 216546_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHI3L1 | 216546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3173
p=0.2753
p=0.214
p=0.455
p=0.8541
p=0.355
p=0.3707
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2